

Arden Monaghan

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EDUCATION

University of Alberta

Sep 2022 – Dec 2026

Bachelor of Science in Computer Science, Specialization

Edmonton, AB

- Cumulative GPA: **3.7** — Major GPA: **3.8**
- **Relevant Coursework:** Data Structures and Algorithms, Introduction to Software Development, Machine Learning 1, Databases & SQL, Statistics 1 & 2, Computer Organization and Architecture, C Programming.
- **Relevant Awards:** Deans Research Award 2025, Jason Lang Scholarship, Deans Honor Roll x2

RELEVANT EXPERIENCE

Software Developer

May 2025 – Present

Society for AI Literacy

Calgary, AB

- Spearheading the full lifecycle development of Syncore EHR Medical technologies, conducting testing at **3 clinics** and achieving a **48% reduction** in administrative overhead for **20+ clinicians**.
- Built **Python-based unit tests** and **Github Actions CI/CD pipeline** to publish **Docker images to AWS ECS** service, reducing team **integration errors** and **deployment time by over 65%**.

ML Research Intern

Jan 2025 – Apr 2025

University of Alberta Electrical and Computer Engineering Faculty

Edmonton, AB

- Implemented a **Camera Calibration system in OpenCV** to estimate **3D hand-joint positions** using ArUco Marker detection and camera-pose estimation, retrieving average offset error of 5% original positions.
- Integrated **SAM technology** using **PyTorch** to correctly compare 3D hand meshes to ground truth 2D masks.
- Fine-tuned model on manually created **dataset of 200 unique hand-object interactions** with applied forces.

Front-End Developer

Nov 2024 – Mar 2025

LitLife Co.

Edmonton, AB

- Facilitated the transition from Firebase to Supabase backend. Mapped **NoSQL to SQL schema** and reintegrated Flutterflow widgets, ensuring **accurate login functionality and profile displays**.
- **Revised user authorization** using **row-level security** systems to reduced harmful database access from unregistered users, increasing application security by over **30%**.

Secretary

Apr 2024 – Mar 2025

Undergraduate Artificial Intelligence Society

Edmonton, AB

- **Lead** partnership with the **UA Innovation Fund** to assist in due diligence on **AI-based startups** using team built scoring matrix which evaluates innovation, market potential, and AI integration strategies.
- Guided over **100 students** per semester in understanding the fundamentals of data cleaning and supervised learning through Scikit-learn workshops.

PROJECTS

[Moodify](#) | *Android Studio, Java, Firebase, Git, Figma*

Jan 2025 – Apr 2025

- **Developed** home feed filtering, user authentication, and personal history **UI/UX** with **Firebase Cloud Storage** for a **Twitter-style Android app**, integrating queries for **displaying posts** from followed users in **real-time**.
- **Debugged and optimized** UI functionality and queries for mood creation, user-following, and user history features using **JUnit** and **Firebase Test Lab**, reducing data display errors by **25%**.

[MentorMatch UAlberta](#) | *Python, Flask, PyQt6, Docker, Ollama*

Apr 2024 – Sept 2024

- Automated the task of matching Mentors and Mentees using an **LLM-integrated algorithm** for the **Science Mentorship Program** at UAlberta, reducing time spent matching from **10 days to 10 minutes**.
- Delivered a PyQt GUI desktop app that integrated Llama3 embeddings with Mentor matching criteria. Enhanced overall program **operations by over 50%** while allowing **student data to remain private**.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, C++, Dart, HTML, CSS, SQL, Swift, RISC-V

Libraries: SkLearn, NetworkX, Pandas, OpenCV, Pytorch, TensorFlow

Technologies/Frameworks: React, Supabase, Firebase, MongoDB, PostgreSQL, QT, Django, FastAPI, SQL Server

Developer Tools: Git, Docker, AWS, Azure